

Capacitive touch

Even though the words ‘capacitive touch’ make us think only about smartphones or tablet screens, there are other areas where its implementation is quite sensitive to the environment and can immediately identify any changes. Some of the lesser-known applications are:

Functional touch-sensitive walls. These can be created by using low-cost carbon based inks in graphics printers for promotional or control purposes. It allows people to interact with these walls, be it at museums, exhibitions or even homes.

Floor and wall integrated capacitive sensors. A tape-like capacitive touch sensor containing carbon ink can be placed across the floor of a room. It sends real-time notifications to the phone whenever a change in capacitance is sensed. This is especially promising for leak detection.

Fig. 5 shows an integrated pressure/temperature sensor that can predict the failure of lithium-Ion batteries when they start producing gas and swell up. These thin, printed sensors can be laminated on top of each other and wrapped around each battery cell to detect temperature and pressure. Such laminated sensor films also contain heating elements to provide an integrated solution.

Disruptive read-out technology for matrix sensors. This is a unique method to obtain signals from a passive matrix array, commonly used for capacitive sensing purposes. So, rather than using voltages to scan each line and then read off the columns, this technology (see Fig. 6) measures variation in the electric current that needs to be supplied to each row or column.

Each element can be simultaneously monitored, while everything else is kept at a steady state; and then the extent of those adjustments is monitored rather than sequentially scanning through the voltage. This solves many issues associated with capacitive touch sensors and also improves signal-to-noise ratio (SNR) and temporal resolution.

For instance, it is impossible to operate a phone with gloves on.



Fig. 5: Integrated pressure/temperature sensors for battery cells in a single printed foil (Credit: InnovationLab)

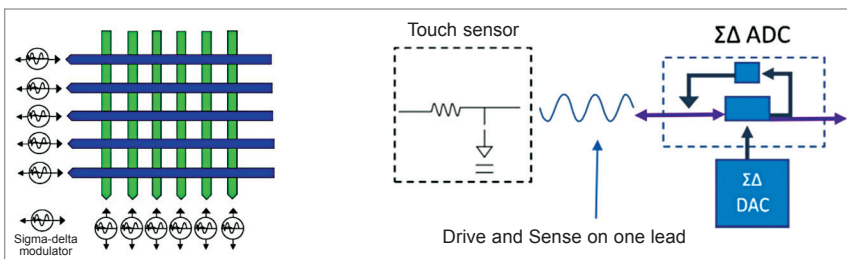


Fig. 6: Disruptive matrix sensor readout technology (Credit: SigmaSense)

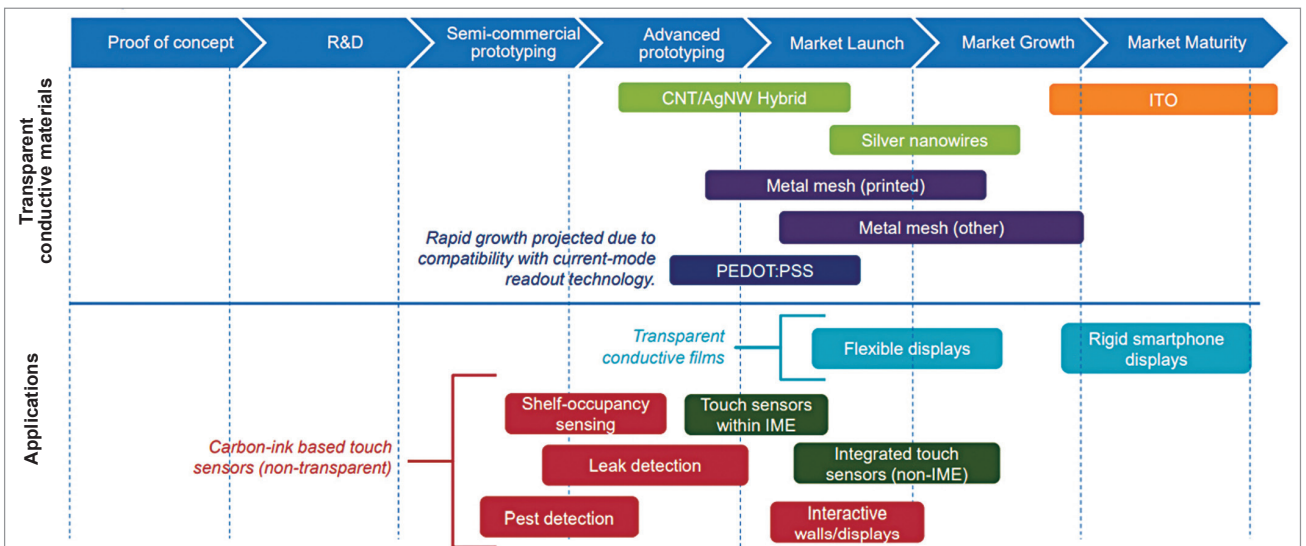


Fig. 7: Readiness level of capacitive touch sensor materials and technologies (Credit: IDTechEx)